

Annotated Bibliography- Competition Lunar Construction Robot, Matthew Geisel

- A. Colwell, J. E., Batiste, S., Horányi, M., Robertson, S., & Sture, S. (2007). Lunar surface: Dust dynamics and Regolith Mechanics. *Reviews of Geophysics*, 45(2).
<https://doi.org/10.1029/2005rg000184>
- a. This article explores the dynamics and electromagnetic properties of lunar regolith (the latter is not really pertinent to this project). The most important findings to this project are the mechanics and qualitative descriptors of regolith. Regolith is found to be very “angular” (sharp) and fine, mostly made up of basalt and glass bombarded by meteorites for billions of years. Due to this, lunar regolith can be dangerous to human health. The physics of regolith excavation are also discussed, as regolith is much easier to excavate at shallow depths (<30 cm) than deeper depths. This describes the type of terrain that the project will exist in.
- B. Mazhar, F., Khan, M. G., & Sällberg, B. (2017). Precise indoor positioning using UWB: A review of methods, algorithms and implementations. *Wireless Personal Communications*, 97(3), 4467–4491. <https://doi.org/10.1007/s11277-017-4734-x>
- a. This article discusses the usage of Ultra wideband (UWB) technology for precise positioning. Advances in UWB technology are discussed, as well as the procedures for deriving position based on time-of-flight measurements. It then discusses the derivation of algorithms to determine the variation of measurements and the accuracy. It then discusses algorithms for how to determine location based on distance measurements. This paper specifies a potential accuracy of over 15 cm, which means UWB is a viable technology for the desired application.
- C. Mermertaş, V. (2004). Optimal design of manipulator with four-bar mechanism. *Mechanism and Machine Theory*, 39(5), 545–554. <https://doi.org/10.1016/j.mechmachtheory.2003.12.002>
- a. This article discusses the design of a four-bar mechanism. These mechanisms are typically used to create parallel motion by creating a moving parallelogram. However, by modifying the length of the links, different motion profiles can be created. This article derives the force-motion relationship of a four-bar mechanism, which is out of the scope of this project. However, these advantages of a four-bar mechanism can be used to store the excavator.
- D. Nagaoka, K., Sawada, K., & Yoshida, K. (2020). Shape effects of wheel grousers on traction performance on Sandy Terrain. *Journal of Terramechanics*, 90, 23–30.
<https://doi.org/10.1016/j.jterra.2019.08.001>
- a. Here, the effect of grouser wheels (wheels that look like gears) on sandy terrain are investigated. A parallel, slanted, V-shaped, and offset V-shape grouser shape was investigated in slip ratio, sinkage, driving torque, and side force tests. The slanted grouser wheel was found to be the highest performing. This is pertinent to the project, as lunar regolith is comparable to sand, so these grousers will likely perform well in lunar regolith if they perform well in sand.
- E. Peng, Y., Qu, D., Zhong, Y., Xie, S., Luo, J., & Gu, J. (2015). The obstacle detection and obstacle avoidance algorithm based on 2-D Lidar. *2015 IEEE International Conference on Information and Automation*. <https://doi.org/10.1109/icinfa.2015.7279550>

- a. This paper discusses obstacle avoidance using a 2D LiDAR sensor. The algorithm presented took 2D readings at an angle so it would intersect the ground in front of the robot. It would take the median to filter noise, split the data into blocks, and cluster based on objects. An avoidance algorithm is then discussed. These algorithms can be implemented in the project to avoid rocks and craters autonomously.
- F. Skonieczny, K., DiGioia, M. E., Barsa, R. L., Wettergreen, D. S., & Whittaker, W. L. (2009). Configuring innovative regolith moving techniques for lunar outposts. *2009 IEEE Aerospace Conference*, 1–11. <https://doi.org/10.1109/aero.2009.4839304>
 - a. This paper discusses some challenges and solutions in constructing permanent human colonies on the Moon. Most relevant to this is the construction of berms. During the Apollo missions, the lifting of the lunar regolith was observed. Therefore, the construction of a berm is useful to prevent this problem. The berm selected for simulation was 2.6 meters tall, had a radius of 25 meters, and a volume of 780 cubic meters. This provides relevance for the project, as berm construction is vital for the Artemis missions.
- G. Vasireddy, K. V., Rose G, M., Suresh, S., M, R., Sharma, S., & Mary Joy, S. (2022). Structural design and implementation of OMNI-directional robot based on Swerve Drive. *2022 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICSES)*, 1–7. <https://doi.org/10.1109/icses55317.2022.9914146>
 - a. This paper discusses the best way to create a swerve drive. A swerve drive is a system where wheels can be rotated about two axes rather than one. This paper creates a small swerve drive robot and discusses the algorithm for controlling it. Overall, the solution is a simpler and easier to implement system that has high speed and maneuverability. This can be used for the project to make object avoidance easier and avoid problems with turning on regolith.
- H. Yang, Z., & Liu, Y. (2008). Quality of trilateration: Confidence based iterative localization. *2008 The 28th International Conference on Distributed Computing Systems*. <https://doi.org/10.1109/icdc.2008.59>
 - a. This article discusses designing algorithms that perform trilateration in imperfect conditions. The paper derives the concept of QoT (Quality of Trilateration) as a more accurate way to determine the zone of probability of location. An equation for the derivation of QoT is discussed. Additionally, the impact of geometry is discussed with respect to QoT. These findings can be used to determine the best placement of the beacons on the field. Additionally, the QoT derivation can be used to derive the location probability field and most likely location.